

JOHN DOMINIQUE L. GONZALES

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ABOUT ME

Dedicated Computer Engineering graduate with a strong foundation in software engineering, clean code practices, and full-lifecycle development. Experienced in building cross-platform mobile applications (React Native, Flutter), full-stack web platforms, and cloud-integrated systems. Proven problem-solver with a solid background in C++, Python, and Java, backed by competitive programming achievements at the regional and national levels. Passionate about writing maintainable, efficient code and contributing to real-world software products in a collaborative engineering environment.

Work Experience

Student Assistant (SPES) – Department of Agrarian Reform Inventory System Developer June 2025 – July 2025

- Designed and developed a centralized web application to digitize and track departmental equipment inventory
- Built a "Softcopy Generation" feature for exporting real-time equipment lists, eliminating manual paperwork
- Architected the transition from physical record-keeping to a responsive digital platform, improving data accessibility and reliability

PROJECTS

Thesis Lead App Designer & Developer

- Designed and engineered an automated, AI-powered Smart Bin system integrating computer vision, IoT edge computing, and real-time cloud data syncing
- Trained a custom YOLO object detection model to classify waste into four distinct streams: paper, plastics, plastic bottles, and unrecyclable materials
- Developed a cross-platform IoT mobile application with real-time fill-level monitoring and automated push notifications via Firebase Cloud Messaging (FCM)
- Implemented Firebase Firestore/Realtime Database to track, aggregate, and visualize daily waste collection metrics and trends per material class
- Programmed an Arduino microcontroller (via Arduino IDE) to orchestrate physical waste segregation mechanisms and manage a compression system for optimal bin capacity

EDUCATION

Marinduque State University

University of Engineering Excellence

Bachelor of Science in Computer Engineering | Aug 2022 - May 2026

- Best Thesis Rank 2**
- Thesis on "AI-powered Smart Bin System"

SKILLS

Technical Skills

Full-Stack Web Development, Cross-Platform Mobile App Development, Cloud Backend Integration (Firebase), Computer Vision & Machine Learning (YOLO), Real-Time Data Streaming, Robotics & Automation Logic

Tools & Frameworks

React Native, Next.js, Node.js, Arduino IDE, PyTorch, MongoDB, Git, MySQL, Postman

Programming Languages

C++, C, C#, Python, Java, Javascript, PHP, Flutter

ADDITIONAL INFORMATION

- Regional C++ Programming Competition** - Champion 2025 & 2nd Place 2026
- National C++ Programming Competition Participant** - 2025
- Civil Service Professional Examination Passer**